

Feasibility as a Parameter: When Comparative Research on Narrow Specialist Professional Populations Can and Cannot Be Executed

Formal Project Report · Draft 1

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Introduction

Comparative international business research rests on a requirement that is rarely stated as a requirement. To compare a phenomenon across countries, a researcher must reach the same population in every country in the design, at the same time, with an instrument that means the same thing in each place. When the population is a general one, this requirement is demanding but routine. When the population is a narrow professional specialty, it becomes something else. This paper is about what that something else is, and about the fact that it can be calculated in advance rather than discovered after the money is spent.

The received wisdom treats sampling difficulty as an execution problem. It belongs to fieldwork, to budget, to the quality of local partners, and it surfaces in published work as a limitation paragraph near the end of the method section. The methodological literature on international survey research is considerably more sophisticated than that, and Harzing, Reiche and Pudelko (2012) document response variation across national frames in detail, but its orientation is managerial. Cross-national variation is a nuisance to be reduced through better translation, better incentives, better contact protocols, and better local relationships. The implicit model is that effort and money convert into responses at a rate that differs by country but is always positive and always improvable.

The gap is that nobody has asked what happens to this picture when the target population is rare. For a population present at a few members per hundred thousand, the conversion of effort into responses is governed less by protocol quality than by arithmetic that is fixed before the first contact is made. More importantly, that arithmetic does not add across national frames. It compounds, and it compounds in one direction only. A design that is marginal in one country is not four times harder in four countries. It can be structurally impossible in four countries while remaining marginal in each one taken separately. That distinction is invisible to a literature that reports sampling as a limitation, because a limitation is by definition something discovered afterward.

The research question this paper proposes to begin answering is therefore: under what conditions is a comparative, multi-country design targeting a narrow specialist professional population feasible at all, and where is the threshold at which a survey design must be abandoned in favour of a different method?

The intended contribution is to move sampling feasibility from the status of a limitation to the status of a parameter. A parameter is estimated before a study is designed and it constrains the design; a limitation is confessed after the study has failed. The claim here is that for narrow populations the relevant quantities are observable in advance. Commercial research panels expose estimated reachable counts on their audience-configuration interfaces before any commitment is made, and published response rates by country are available. Those two inputs, combined with an eligibility screen, determine feasibility. A secondary contribution follows from the first and is the more useful one for practice. Where the arithmetic rules out a survey it does not rule out the study. The same twenty reachable participants who cannot sustain a comparative survey can sustain a phenomenological design. The threshold at which one method fails and another becomes appropriate is locatable before resources are committed, and locating it is a service to the field rather than an admission about one project.

This is a contribution to an existing conversation rather than a new one. The regionalization literature established that firms described as global are substantially home-region bound (Rugman & Verbeke, 2004), and that ventures described as born global may be better described as born regional (Lopez, Kundu, & Ciravegna, 2009). This paper extends the same suspicion, that the label and the reality diverge, from the firms under study to the research infrastructure used to study them. If commercial research panels are themselves firms, and firms are regionally bound, then the infrastructure on which comparative research depends should be regionally bound in the same way and for the same reasons. That is an empirical question and it has not been asked.

Literature Review

Four streams bear on the question. The first documents the phenomenon without modelling it. The second supplies the theoretical mechanism for why research infrastructure should be regionally bounded. The third supplies the levels of analysis at which feasibility is determined. The fourth supplies the tension between standardization and response that sits at the centre of any comparative instrument.

International survey research and the management of response variation

Harzing, Reiche and Pudenko (2012) provide the most detailed available account of what actually happens when a single instrument is fielded across national frames. Their project response rates range from four percent in China to roughly fifteen percent in Spain, with a figure of forty-seven percent in Korea obtained by telephone through a survey company rather than by the instrument used elsewhere. The orientation of this work is practical. It tells a researcher what to expect and what to do about it.

Two features of that dataset matter more than the paper's own emphasis suggests. The first is the spread. A four-fold difference between the best and worst frame is not noise around a mean; it is a difference in kind, and any design that requires a threshold in every frame is

governed by the worst of them. The second is the Korean observation. A change of mode moved the response rate by roughly five times more than the entire spread attributable to country. If that pattern generalises, then the variable the comparative literature treats as central, national context, is dominated by a variable it treats as incidental, instrument mode. The present paper takes that inversion seriously enough to hypothesise it.

Regionalization and the gap between the label and the reality

Rugman and Verbeke (2004) examined the sales of the largest multinationals and found that the overwhelming majority were home-region oriented rather than global. The theoretical mechanism is not simply distance. Firm-specific advantages developed in a home region transfer within that region at low cost because the supporting institutions, customer expectations and distribution structures are similar, and transfer across regions only by being rebuilt. The label global describes an aspiration; the sales data describe a region.

Lopez, Kundu and Ciravegna (2009) applied the same test to firms that appear international from inception. Their Costa Rican software cases are international from day one in the sense the born-global literature intends, yet their activity concentrates regionally. The contribution is the demonstration that a firm can satisfy the formal definition of a category while failing the substantive one, and that the discrepancy is detectable only by measuring where the activity actually lands rather than where the firm says it operates.

The extension proposed here is direct. A commercial research panel is a firm. It recruits its membership through channels that are themselves institutionally embedded: professional associations, employment platforms, advertising networks and payment infrastructures that differ by region. There is no reason to expect the resulting membership to be regionally neutral, and considerable reason grounded in this literature to expect it to be home-region concentrated. For a general population the concentration may be immaterial because the population is abundant everywhere. For a specialty present at a few per hundred thousand, a

modest regional skew in recruitment is the difference between a reachable frame and an unreachable one.

Levels of analysis and the agency of local units

Meyer, Li and Schotter (2020) argue that the subsidiary is not an instrument of headquarters but an actor with its own agency, embedded simultaneously in the internal network of the firm and in an external national context, and that research on it must be explicitly multi-level and dynamic. Their agenda is a corrective to designs that treat the country as the only level that matters.

The parallel to research design is exact and has not been drawn. Feasibility is determined at several levels at once: at the population level by prevalence, at the infrastructure level by panel coverage, at the instrument level by mode and length, and at the design level by the strictness of the eligibility screen. These levels are not nested cleanly and they are not interchangeable. A researcher who attributes a failed frame to the country has committed the error Meyer and colleagues describe, which is to assign to one level a result produced at another. The Korean telephone datum is precisely such a case.

Control, coordination, and the standardization tension

Zeng and colleagues (2023) review the mechanisms by which multinationals control and coordinate dispersed units, and separate the two. Control is the imposition of a common standard from the centre; coordination is the alignment of units that retain discretion. The two are not the same instrument and they trade off, because tightening the first reduces the local adaptation on which the second depends.

Comparative survey research faces this tradeoff in an unusually pure form. Measurement equivalence requires that the instrument be as nearly identical as possible across frames, which is control. Achieving response requires adaptation to local norms of length, channel, incentive, sponsorship and language register, which is coordination. Every increment of standardization

purchased for the sake of comparability is paid for in response, and the payment is heaviest in exactly the frames that were weakest to begin with. The literature treats equivalence and response as separate methodological concerns. They are the two sides of one tradeoff, and for a narrow population the tradeoff binds.

The recurring lesson: measuring one thing while believing another was measured

A final stream is relevant less for its content than for its shape. Contractor, Kundu and Hsu (2003) resolved three decades of contradictory findings on multinationality and performance by showing that the field had been sampling different segments of a single sigmoid curve and reporting each segment as a finding about the whole. Mezias (2002) made progress on the liability of foreignness only by abandoning aggregate performance measures, which confound advantages with disadvantages, in favour of labour lawsuit judgments, which isolate one disadvantage. Ghemawat (2001) showed that country portfolio analysis measures market size while being used as though it measured opportunity, and that firms which solve geographic distance routinely conclude they have solved distance.

In each case the obstacle was not the theory but what the field was able to measure, and on whom. This paper takes that problem as its subject rather than as a limitation paragraph. The question of who can actually be reached, in which frames, and what follows for the design, is treated here as a first-order research question in international business methodology.

Hypotheses Development

Five hypotheses follow. The first is analytic and states the structure of the problem. The remaining four are empirical and each draws its theoretical logic from one of the streams above.

H1. The binding-frame hypothesis

In a comparative design that requires a minimum usable sample in every national frame, joint feasibility is determined by the least feasible frame rather than by the mean across frames. Adding a national frame to such a design weakly decreases joint feasibility and can never increase it.

The theoretical logic does not depend on any citation and should be clear without one. A comparative claim is a conjunction. To say that a relationship differs between Spain and China is to assert something about Spain and something about China, and the assertion fails if either component fails. The design therefore requires that every frame clear its threshold simultaneously. An average cannot satisfy a conjunction. A design with one excellent frame and one hopeless frame has a respectable mean and no comparison.

The consequence is asymmetric in a way that is easy to miss. Because the requirement is set by the minimum, and because a set's minimum can only fall or stay level when an element is added, every additional country weakly worsens the design and no additional country can improve it. Breadth, which the field treats as a virtue in comparative work, is therefore purchased at a price that rises faster than the number of frames. This is not a claim about difficulty. It is a claim about structure, and it means that the common practice of describing a design as more ambitious because it covers more countries misstates what has been done: the design has not become more ambitious, it has become more fragile.

Worked against the empirical inputs described in the method section, the magnitude is large. A single-frame design in the most favourable country requires roughly nineteen million panel members to yield thirty usable responses. Adding four more countries, one of which is the least favourable, raises the requirement in every frame to roughly seventy-two million while the mean response rate across the design falls only from about fifteen percent to about nine percent. The mean is nearly uninformative about the design's viability, and reporting it is

reporting the wrong statistic.

Insert Table 1 About Here

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H2. The regional bounding hypothesis

The observable prevalence of a narrow specialist professional population on a commercial research panel is higher within the panel provider's home region than outside it, controlling for the size of the national frame.

The theoretical logic is the mechanism Rugman and Verbeke identify, applied to a firm whose product is access to people. A panel provider builds its membership through recruitment channels that are institutionally specific: professional associations, employment platforms, advertising markets, payment rails and reputational networks. Each of these is denser and cheaper to work within the region where the provider originated, because the provider's own advantages in that region are the advantages Rugman and Verbeke describe, and they do not transfer without being rebuilt.

The reason this should bite harder for specialists than for general populations is that specialist recruitment runs through narrower channels. A general consumer panel can be assembled almost anywhere through broad advertising. A panel containing experienced internal auditors is assembled through professional bodies and industry networks whose reach is national or regional by construction. The rarer the specialty, the more the panel depends on exactly the channels least likely to have been rebuilt outside the home region. The prediction is therefore not merely that coverage is uneven but that its unevenness increases as the target population narrows.

H3. The born-regional design hypothesis

Published studies that describe themselves as cross-national achieve national coverage that is more regionally concentrated than their stated design implies, and the discrepancy increases with the narrowness of the target population.

The logic is the one Lopez, Kundu and Ciravegna used on born globals, transposed. A study satisfies the formal definition of cross-national by including frames from several countries. Whether it satisfies the substantive definition depends on where the achieved responses actually land, which is a different measurement and is rarely reported with enough granularity to check.

The mechanism connecting this to H2 is straightforward. If panel coverage is regionally bounded, then achieved responses will concentrate in the frames where coverage is best, which will be the frames within the provider's home region. Researchers who set a global target and accept the responses that arrive will produce a sample that is formally multinational and substantively regional, without any decision having been made to that effect. The concentration is a property of the infrastructure, not of the research design, which is precisely why it goes unremarked. For a narrow population the effect should be larger, because the weakest frames contribute close to nothing and the sample collapses onto the strongest ones.

H4. The mode dominance hypothesis

Variance in achieved response attributable to data-collection mode exceeds variance attributable to country.

The logic comes from Meyer, Li and Schotter's insistence that outcomes be assigned to the level that produced them. Response is an outcome of an encounter between a person and a request. The properties of that encounter, whether the request arrives by telephone from a human being or by email from a platform, how long it claims to take, who appears to be asking, are properties of the instrument, not of the country. National context shapes the

encounter, but it does so through norms that are themselves mode-specific: the norm governing an unsolicited telephone call is not a weaker or stronger version of the norm governing an unsolicited email, it is a different norm.

The reason to expect mode to dominate rather than merely to matter is that mode changes the cost structure of compliance for the respondent, while country changes only the disposition toward it. A telephone request from a person imposes a social cost on refusal that an email does not impose in any country. The single available datum is consistent with this: the forty-seven percent obtained by telephone in Korea exceeds the entire country spread in the same project by roughly a factor of five. One observation is not evidence, which is why this is a hypothesis and not a finding, but it is the observation that makes the hypothesis worth testing. If it holds, a substantial part of what the comparative literature has attributed to national context is an artefact of instrument selection.

H5. The standardization tension hypothesis

The degree of instrument standardization across national frames is negatively associated with achieved response rate, and the association is stronger in frames with lower baseline response.

The logic is Zeng and colleagues' separation of control from coordination. Standardization is control: a common instrument imposed from the centre so that responses are comparable. Response depends on coordination: adaptation of length, channel, sponsorship, incentive and language register to what each frame will actually tolerate. These draw on the same instrument and pull it in opposite directions. Every element held constant for the sake of equivalence is an element that cannot be adapted for the sake of response.

The interaction is the part that matters for design. In a strong frame, standardization costs response that the frame can afford to lose. In a weak frame, the same standardization is applied to a frame that had no margin, and it is the weak frames that bind the design under H1. Standardization therefore damages a comparative design at precisely the point where the

design is already most vulnerable, which means the equivalence-response tradeoff is not linear in its consequences and cannot be managed by trading a uniform amount of one for the other.

Insert Table 3 About Here

Methodology

Study context

The empirical setting is the set of commercial research panels used in academic and commercial survey work, and the specialist professional occupations that comparative business research targets. The setting is unusual in one respect that makes the study possible. Panel providers expose an audience-configuration interface that returns an estimated reachable count once screening criteria are entered, before any commitment is made and at no cost. The quantity this study needs as its dependent variable is therefore observable in advance and is quotable, rather than having to be inferred from completed studies. This is what makes feasibility a parameter rather than a limitation, and it is the practical basis of the whole project.

The unit of analysis is the country-panel-occupation cell: one specialist occupation, on one panel provider, in one national frame. The sampling plan is to populate a matrix of at least six specialist occupations across at least eight national frames on at least three panel providers of differing regional origin, which gives a target of roughly one hundred and forty-four cells. Panel providers are selected to vary on home region deliberately, because H2 is unidentifiable if every provider originates in the same place.

The author's own study supplies one fully documented validating case. A panel of 334,976 members returned approximately twenty eligible participants under the study's screening criteria, a prevalence of about six per hundred thousand, and of twenty-three raw responses four survived screening. Cost per usable response is documented. Any model proposed here must reproduce that case from its own inputs, and a model that cannot do so is rejected.

Variable measures

The dependent variable is reachable usable sample, defined as the estimated number of responses that would survive eligibility screening and data-quality exclusion in a given cell. It is computed as the product of frame size, observed prevalence, expected response rate and screen survival. Where achieved response rates are observable from published studies, achieved response rate is used as a second dependent variable for H4 and H5.

The independent variables follow the hypotheses. For H2, region concordance is a binary indicator of whether the national frame falls within the panel provider's home region, with region defined on the triad classification used by Rugman and Verbeke. For H3, coverage discrepancy is the difference between the regional concentration implied by a study's stated design and the concentration of its achieved responses, computed as a Herfindahl-style index over regions. For H4, mode is a categorical variable distinguishing telephone, email or platform panel, face to face, and mixed. For H5, standardization is an index constructed over whether the instrument held constant its item wording, length, incentive structure, sponsorship presentation and contact protocol across frames.

Control variables are population prevalence, frame size, eligibility strictness measured as the number of conjunctive screening criteria, incentive value expressed in purchasing power parity, field period length, and year of fielding. Eligibility strictness is included because it is the one term in the feasibility identity that is under the researcher's control, and any finding that does not separate it from the terms that are not under control would be uninterpretable.

Statistical method

H1 is an analytic claim rather than an inferential one and is treated accordingly. It is established by demonstration and probed by sensitivity analysis over the full plausible range of prevalence, response and screen-survival values, reporting the region of the parameter space in which a comparative design remains viable. This is the appropriate treatment because the claim is about the structure of a conjunction and would not be made more true by a p-value. Reporting a significance test for it would misrepresent a deductive result as an empirical one.

H2 through H5 are tested with hierarchical linear models, with cells nested within national frames and frames nested within regions, and with random intercepts at the frame and provider levels. The nesting is not a statistical convenience; it is the substantive structure that Meyer and colleagues argue must be modelled explicitly, and ignoring it would attribute to countries the variance that belongs to providers. For H5 the standardization-by-baseline interaction is the term of interest, and the hypothesis is supported only if the interaction is significant in the predicted direction, not merely the main effect.

Where the instrument yields a scale rather than a single observable, reliability is established before use. The standardization index is the only constructed scale in the design; its internal consistency will be reported, and the individual components will be reported separately as well, because an index that conceals a divergent component would hide exactly the tradeoff the hypothesis concerns.

No data have been collected for this project. The design above is what the project proposes, and the preliminary evidence reported in the discussion comes from the author's prior study and from published response rates, not from fieldwork conducted for this paper.

Discussion and Conclusions

The expected pattern across the five hypotheses is a single one stated five ways: that the feasibility of comparative research on rare populations is determined by structural properties of the research infrastructure that are observable in advance, and that the profession's habit of treating these properties as execution problems has caused them to be discovered late and reported as limitations.

H1 is expected to hold by construction, and the contribution lies in its magnitude rather than its direction. The demonstration that the mean response rate across a design is nearly uninformative about that design's viability, while the minimum is decisive, has an immediate consequence for how comparative work should be reported. Studies currently report an overall response rate. Under H1 that number is the wrong summary statistic and a per-frame minimum should be reported alongside it.

H2, if supported, would establish that the regionalization thesis applies to research infrastructure and not only to the firms that infrastructure is used to study. This is the most consequential of the four empirical hypotheses, because it implies that comparative findings in the field carry a regional signature imported from the panels rather than from the phenomena. If it is not supported, the more likely explanation is that the major providers have converged on similar global recruitment channels, which would itself be worth documenting and would bound the claim usefully.

H3, if supported, would show that the divergence between stated and achieved coverage is systematic rather than incidental, and that it scales with population narrowness. The practical yield is a reporting standard: achieved coverage by frame, reported as a matter of course, in the way that response rates already are.

H4 is the hypothesis most likely to be wrong and the most valuable if it is right. It rests on a single observation from one study, which is not evidence. If mode does dominate country, then

a portion of the cross-national variation the field has interpreted substantively is methodological, and comparative designs should be matched on mode before they are compared on context. If it does not hold, the finding still disciplines the literature by establishing that the Korean datum was idiosyncratic, which is worth knowing because that datum currently sits in a widely cited source without qualification.

H5, if supported, would identify the equivalence-response tradeoff as a design decision with an interaction rather than a uniform cost, and would show that standardization is most damaging where the design is already weakest. The prescription that follows is not to abandon equivalence but to concentrate adaptation in the binding frames, which is a specific and testable recommendation rather than general advice to balance the two.

Theoretical contributions

The principal theoretical contribution is to reposition sampling feasibility as a parameter of a comparative design rather than a limitation of a completed one. This changes what is knowable and when. A parameter is estimated in advance and constrains the design space; a limitation is a retrospective account of why a design did not work. The paper supplies the identity that links the observable quantities to reachable sample and demonstrates that the quantities are available before commitment.

The second contribution extends the regionalization argument to a new object. Rugman and Verbeke, and Lopez, Kundu and Ciravegna, established that the gap between a firm's stated geographic scope and its realised scope is systematic. This paper proposes that the same gap exists for research infrastructure and, by extension, for the studies built on it. That is an instance of using an existing theory to look at a new phenomenon, which is one of the routes to contribution the seminar identifies.

The third contribution is a levels-of-analysis correction. By separating population, infrastructure, instrument and design as distinct determinants of feasibility, the framework

makes it possible to state which level produced a failed frame. The current convention, which attributes such failures to the country, commits the error Meyer and colleagues warn against and does so systematically.

Managerial and practical contributions

For researchers the yield is a pre-commitment test. Before a comparative design is funded, the reachable sample in the weakest intended frame can be computed from quantities that cost nothing to obtain. A design that fails this test fails before money is spent rather than eighteen months afterward. The author's own case is the argument for why this matters: the constraint was discovered after fielding, and the cost of that discovery was the study.

The second practical yield is the method-switch threshold. The framework identifies the point at which a population that cannot sustain a survey can still sustain an interview-based design, and it locates that point in advance. This reframes a class of research problems currently treated as failures. A population of twenty reachable participants is a failed survey and a well-powered phenomenological study, and knowing which of those two a project is before it begins is the difference between a contribution and a write-off.

For organisations commissioning multi-country research the implication is a procurement one. A vendor proposal covering eight markets should be evaluated on its weakest market, and the regional origin of the panel should be treated as material information about coverage rather than as a corporate detail.

Limitations

The feasibility model holds prevalence constant across countries, and it almost certainly is not constant. No cross-national prevalence data for specialist professional populations currently exists, which is why obtaining it is the empirical contribution the project proposes rather than an assumption it can rely on. Until that data exists, the model's cross-national figures should be read as illustrating a structure, not as estimating a quantity.

The response rates used come from a single study, one instrument and one period. They are the best available and they are not sufficient. Treating them as a general country characteristic would commit the level-of-analysis error the paper itself identifies, and H4 exists in part to test whether that error is being made.

The model treats national response rates as independent. They are not; frames within a region share institutional conditions and will correlate, which means the true joint feasibility of a multi-frame design is worse than an independence assumption implies in some configurations and better in others. Estimating the correlation structure requires the data the project proposes to collect.

Finally, panel audience-configuration estimates are vendor-supplied and are not audited. They are the same numbers researchers currently rely on when commissioning work, so using them does not introduce a new exposure, but a systematic comparison of estimated against achieved counts is a necessary validation step and is not yet part of the design.

Conclusion

Comparative international business research has treated the question of who can be reached as a matter of execution. For general populations that treatment is harmless. For narrow specialist populations it is not, because the arithmetic that governs reachability compounds across national frames in one direction and is decided by the weakest frame rather than the average one. The quantities involved are observable before a study is fielded. Making them part of the design, rather than part of the confession, is a small change in practice and a substantial change in what the field can claim to have compared.

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Tables and Figures

Table 1. Panel frame required for 30 usable responses, by country

Country	Response rate	Panel frame required
China	4.0%	72,229,200
United Kingdom	5.2%	55,560,923
France	6.6%	43,775,273
Japan	10.4%	27,780,462
Germany	11.1%	26,028,541
Nordic countries	11.3%	25,567,858
Australia / New Zealand	12.7%	22,749,354
Spain	15.4%	18,760,831
Korea (telephone)	47.0%	6,147,166

Note. Response rates from Harzing, Reiche and Pudenko (2012), Illustration 7. Prevalence of 5.97 per 100,000 and screen survival of 0.174 are from the author's own study. Korea's rate was obtained by telephone through a survey company rather than by the instrument used in the other frames and is reported separately for that reason.

Table 2. What a design requires as national frames are added

Design	Mean response rate	Binding rate	Frame required in EACH country
Spain	15.4%	15.4%	18,760,831
+ Germany	13.2%	11.1%	26,028,541
+ Japan	12.3%	10.4%	27,780,462
+ United Kingdom	10.5%	5.2%	55,560,923
+ China	9.2%	4.0%	72,229,200

Note. Target is 30 usable responses in every frame in the design. The binding rate is the minimum response rate among the frames included. The mean falls by roughly a third while the requirement rises by roughly a factor of four, which is the asymmetry H1 predicts.

Table 3. Hypotheses, theoretical source, and test

	Hypothesis	Theoretical source	Test
H1	Joint feasibility is set by the least feasible frame, not the mean; adding a frame weakly decreases it	Analytic; conjunction structure	Demonstration plus sensitivity analysis over the parameter space
H2	Specialist prevalence on a panel is higher inside the provider's home region	Rugman & Verbeke (2004)	Hierarchical model; region concordance indicator
H3	Studies described as cross-national achieve regionally concentrated coverage, increasing with population narrowness	Lopez, Kundu & Ciravegna (2009)	Coverage discrepancy index across published studies
H4	Mode explains more variance in response than country does	Meyer, Li & Schotter (2020)	Variance decomposition across levels
H5	Standardization is negatively associated with response, more so in weak frames	Zeng et al. (2023)	Standardization by baseline-response interaction

Figure 1. The binding-frame effect

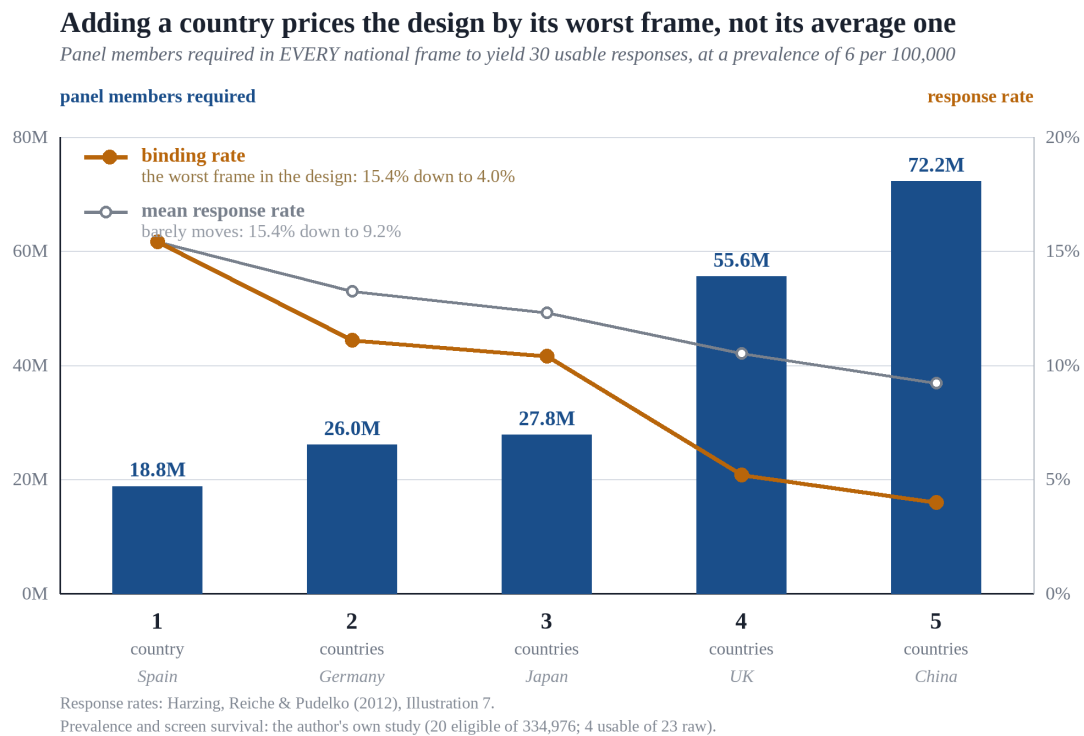


Figure 1. Adding a national frame prices the design by its worst frame, not its average one. Response rates from Harzing et al. (2012); prevalence and screen survival from the author's own study.

Disclosure of Artificial Intelligence Use

Generative artificial intelligence (Claude, Anthropic) was used to organise this draft, to format it to the required structure, to produce Figure 1 from the author's own model code, and to check the prose for clarity and grammar. The research question, the theoretical argument, the hypotheses and their justification, and the interpretation of the empirical inputs are the author's own. The figures reported derive from the author's own study records and from published sources cited in the text. No confidential, identifiable, or restricted data was provided to any artificial intelligence system. This disclosure is made in accordance with the FIU University Graduate School policy on artificial intelligence use and the course syllabus.